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REVIEW ARTICLE



Perspectives on the use of machine learning for ADME prediction at AstraZeneca

Erik Gawehn^a, Nigel Greene^b , Filip Miljković^c, Olga Obrezanova^d, Vigneshwari Subramanian^a, Maria-Anna Trapotsi^d and Susanne Winiwarter^e

^aImaging and Data Analytics, Clinical Pharmacology & Safety Sciences, R&D, AstraZeneca, Gothenburg, Sweden; ^bImaging and Data Analytics, Clinical Pharmacology & Safety Sciences, R&D, AstraZeneca, Waltham, MA, USA; ^cMedicinal Chemistry, Research and Early Development, Cardiovascular, Renal and Metabolism (CVRM), BioPharmaceuticals R&D, AstraZeneca, Gothenburg, Sweden; ^dImaging and Data Analytics, Clinical Pharmacology & Safety Sciences, R&D, AstraZeneca, Cambridge, United Kingdom; ^eDrug Metabolism and Pharmacokinetics, Research and Early Development, Cardiovascular, Renal and Metabolism (CVRM), BioPharmaceuticals R&D, AstraZeneca, Gothenburg, Sweden

ABSTRACT

1. A drug's pharmacokinetic (PK) profile will determine its dose and the frequency of administration as well as the likelihood of observing any adverse drug reactions.
2. It is [REDACTED]
3. In this paper, we describe the approaches used within AstraZeneca to improve our ability of predicting the preclinical and human pharmacokinetic profiles of novel molecules using machine learning and artificial intelligence.
4. We will show how combining chemical structure-based approaches with experimentally derived properties enables improved predictions of *in vivo* pharmacokinetics and can be extended to molecules that go beyond the classical Lipinski's rule-of-five space.
5. We will also discuss how combining these *in vitro* and *in vivo* predictive models could ultimately improve our ability to predict the human outcome at the point of chemical design.

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Introduction

Drug discovery and development is still an extremely expensive and complex process with an average time of 10–15 years to get a new drug to the market and an estimated failure rate of ~90% even after a drug has made it into clinical trials (Sun et al. 2022). The two leading causes are lack of clinical efficacy (40–50%) and unmanageable toxicity (30%) based on an analysis of the reasons for clinical failure between 2016–2018 (Sun et al. 2022). Many of these potential drug candidates originate outside of the large pharmaceutical industry, more concretely from academic and small biotechnology companies (Schuhmacher et al. 2023).

As a direct result there has been a push towards automation and increasingly digitised workflows including the adoption of various forms of machine learning (ML) methods into the drug development process that have transformed how drugs are developed in more recent times (Schneider 2018; Jiménez-Luna et al. 2021) and have contributed to the increased productivity of research and development at AstraZeneca (Morgan et al. 2018).

One area profiting from the adoption of these new paradigms and tools is the determination of a drug's pharmacokinetic and pharmacodynamic (PK/PD) profile (Morgan et al.

2018). These umbrella terms describe 'what the body does to the drug and what the drug does to the body', respectively. Pharmacokinetics (PK) is characterised in terms of four phases a substance goes through after entering the body: Absorption, Distribution, Metabolism and Elimination (ADME). PK issues have previously been the leading cause of drug terminations during clinical trials, however, the advancements in the last decade in early PK/PD profiling have reduced those risks and now other factors like efficacy and toxicity take the top spots today (Waring et al. 2015; Morgan et al. 2018).

During the drug discovery process, it is important to be able to predict human PK parameters to establish a suitable dose for clinical trials. Hence, various *in vitro* assays are routinely used to characterise a drug's ADME profile during the lead optimisation stage of drug discovery (Pillai et al. 2022). These are typically followed by preclinical *in vivo* studies to further characterise compound's PK parameters with the ultimate goal being to extrapolate to humans. However, the ability to accurately predict the experimental results of these studies from the chemical structure offers the advantages that only compounds with the desired characteristics are prioritised for synthesis and the experimental data becomes confirmatory rather than exploratory. Similarly, having the

ability to accurately predict human PK parameters at the point of compound design would provide significant improvements in the efficiency of the drug discovery process.

To develop such *in silico* models, sufficient quantity and quality of data is required. The availability of high-throughput assays for PK-related properties such as the logD, solubility, cell permeability and human liver microsomal clearance, result in large quantities of data being available for model training. However, *in vivo* PK experiments are more complex and expensive, and hence there is generally less data resulting in *in silico* models with potentially smaller applicability domains and lack of new data to prospectively validate the existing models.

PK parameters are inter-related but can be grouped as described by Davies et al. (Davies et al. 2020); into primary and secondary PK parameters. Examples of primary PK parameters are clearance (CL) that is related to the drug's elimination process and is indicative of the rate at which a drug is eliminated from the body (Pantaleão et al. 2022); volume of distribution (Vd) that is a measure of the relative affinity of a drug for plasma and tissues, and bioavailability (F) that is defined as the fraction of an extravascular administered drug to reach systemic circulation (Davies et al. 2020). Primary PK parameters are mechanistic parameters related to the fundamental ADME processes. Vd and CL require intravenous (IV) administration, F requires also extravascular, e.g. oral administration (PO).

Secondary PK parameters, such as maximum concentration of a drug in the systemic circulation (C_{max}) at time T_{max} , half-life ($t_{1/2}$) or area under the curve (AUC), can be derived from the primary (predicted) PK parameters but also from the observed concentration time curve, irrespective of the route of administration (Figure 1). A drug's PK properties influence the therapeutic window between the minimally effective concentration (MEC) at which a drug has a biological effect and the maximally tolerable concentration (MTC) at which a compound becomes toxic.

In the following, we will discuss some of the recent work done on using ML to predict *in vitro* ADME and *in vivo* PK parameters, however, for a more extensive overview we would refer the reader to recent reviews by Oprisiu and Winiwarter (Oprisiu and Winiwarter 2020) and by Obrezanova (Obrezanova 2023). We will also discuss how AstraZeneca has used ML models to mimic the experimental testing cascade to improve the prediction of human PK properties at the point of chemical design.

Tiered approach to prediction of human pharmacokinetics

For the last several decades, pharmaceutical companies have used physicochemical properties and *in vitro* assays to estimate the likely human PK parameters of novel molecules (Figure 2). This is typically followed by *in vivo* studies in preclinical species that are then used to further refine these estimates of the human PK profiles. Traditionally, ML models based on chemical structure have been developed on the relatively large sets of data from these *in vitro* assays to help with the design and selection of molecules with more desirable profiles.

Predicting physicochemical properties

Lipophilicity models

Lipophilicity is the most common parameter in drug discovery, as it influences a wide range of drug metabolism and pharmacokinetics (DMPK) properties, compound potencies and safety-related properties. At AstraZeneca, lipophilicity is assessed based on a compound's distribution coefficient between octanol and water at pH 7.4, using the shake-flask approach ($\log D_{7.4}$) and a high-performance liquid chromatography (HPLC) approach (ChromLogD) (Wan and Holmen 2009). The shake-flask method is conventionally used for the

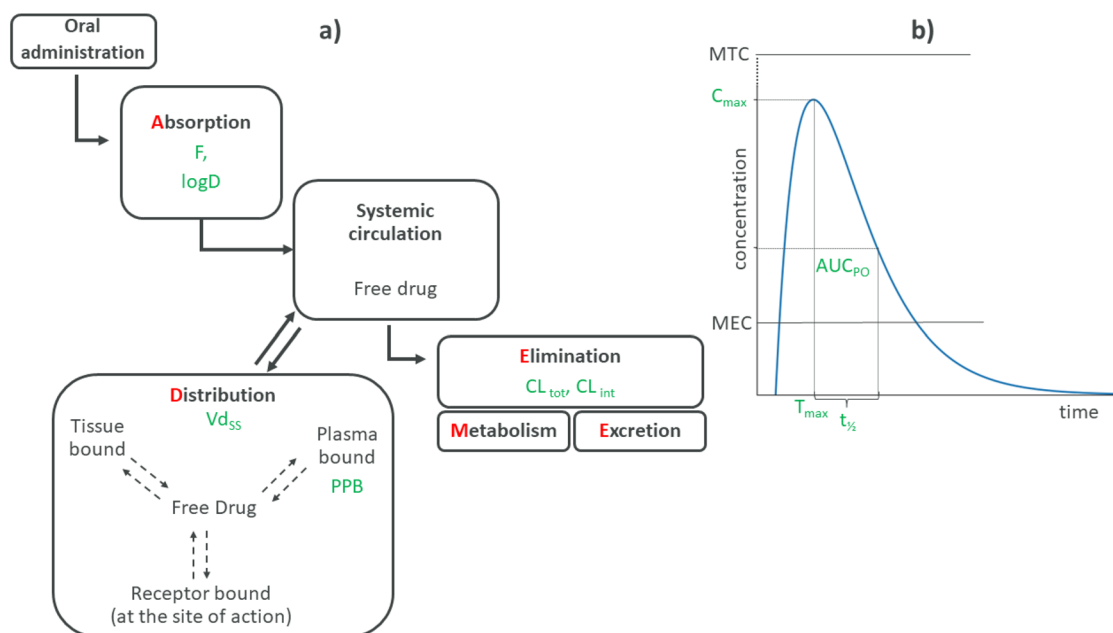


Figure 1. Schematic overview of phases in a drug's disposition after oral administration with a selection of associated PK properties (a) and a concentration-time curve for a single, oral drug dose (b).

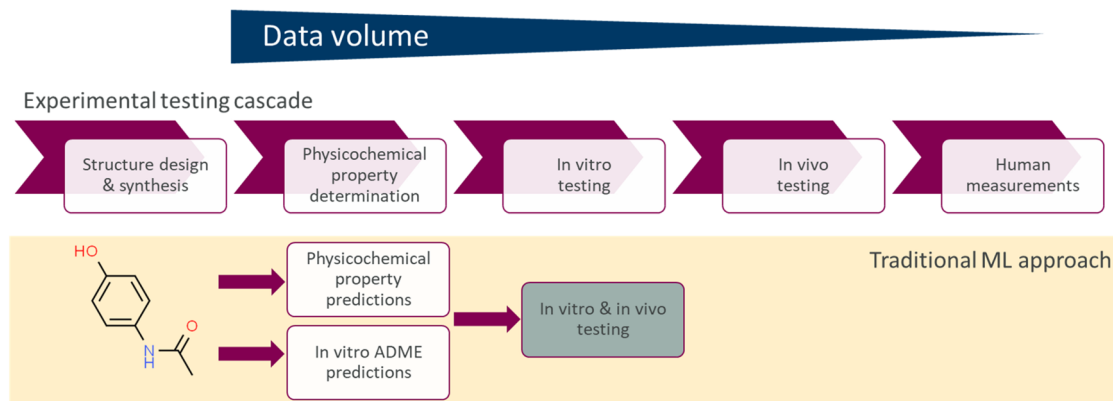


Figure 2. Machine learning in the prediction of ADME properties.

majority of the compounds, whereas HPLC is primarily used for compounds whose expected lipophilicities are over 4 (e.g. PROteolysis TARgeting Chimaeras or PROTACs). With the huge wealth of data generated over the years, we have developed ML models for predicting both shake-flask and chromatographic logD. Previously, the combination of signature descriptors (Faulon et al. 2003) and support vector machine (SVM) algorithm (Cortes and Vapnik 1995) was found beneficial for many molecular property prediction tasks (Alvarsson et al. 2014), and since CPSign software (CPSign) conveniently combines both within a single modelling framework (including the conformal predictor component) it was selected as a preferred *in silico* tool for many AstraZeneca models. The lipophilicity models rely on signature descriptors, which are vector representations of atomic signatures corresponding to canonical representations of atom environments up to a pre-defined height (h) (Faulon et al. 2003). The models are trained using the SVM algorithm, which is a supervised approach that constructs a hyperplane with maximal margin to separate two compound classes (classification) or that minimises the difference between an observed and a predicted numerical value (regression; support vector regression variant) (Cortes and Vapnik 1995). Further, to provide uncertainty estimates, CPSign's conformal prediction framework was adopted. It serves as an alternative approach to applicability domain estimation where each compound query is compared to the training set with respect to both the descriptor space and the prediction outcome. Given a particular confidence value, a prediction interval rather than a single regression value is provided with each compound prediction (Norinder et al. 2014; CPSign). In summary, the *in silico* logD models offer predicted values along with uncertainties estimated at 70% confidence intervals. These models are frequently used to prioritise compounds with the right lipophilic profiles in the pre-clinical phases of the projects.

ePSA model

Exposed Polar Surface Area (ePSA), an indicator of a molecule's permeability across the membrane is measured using supercritical fluid chromatography technique (Goetz et al. 2014). Polar interactions with stationary phase in an apolar solvent are captured as retention times, which are then

transformed to ePSA values provided as Å². With the ePSA data being available for ~17,000 small molecules and PROTACs, we have built a random forest model (Breiman 2001) based on OESelma 2D descriptors (Linusson et al. 2001) and Extended connectivity fingerprints (Rogers and Hahn 2010). The model is implemented using the scikit-learn module (Pedregosa et al. 2011) of Python and the output corresponds to the predicted values together with the uncertainties computed considering the standard deviations of the predictions from individual trees.

Solubility models

Solubility is yet another crucial property in drug discovery that has an effect on bioavailability, absorption, drug dosing, and formulation (Kerns et al. 2008). Typical experimental solubility measurements in AstraZeneca are based on compound Dimethyl Sulfoxide (DMSO) solutions, where DMSO is evaporated before aqueous solubility at pH 7.4 is measured in a shake-flask approach (Wan and Holmen 2009). The best use of ~150,000 measured solubility values comes from our *in silico* modelling efforts to build a SVM (Cortes and Vapnik 1995) regression model using signature descriptors (Faulon et al. 2003). Similar to lipophilicity models, the implementation was done through CPSign conformal prediction framework (Norinder et al. 2014; CPSign) and predicted values/uncertainties at 70% confidence interval are provided to users. High experimental variability makes it really challenging to generate a predictive regression model. To complement the existing regression model, we generated a conformal classification model that classifies compounds with solubilities below or above 50 µM. The classification model was also built through the CPSign framework and the p-values corresponding to each class label (high: >50 µM and low: <50 µM) were used to assess the confidence and credibility of the predictions. Both regression and classification models are used internally in compound prioritizations, depending on the needs of the project. The classification model was also built through the CPSign framework and the p-values corresponding to each class label (high: >50 µM and low: <50 µM) were used to assess the confidence and credibility of the predictions. Both regression and classification models are used internally in compound prioritizations, depending on the needs of the project.

Performances of physicochemical models

We find both the lipophilicity and the ePSA models are usually very reliable, with test set performances with a coefficient of determination (R^2) greater than 0.8 and a root mean squared error (RMSE) less than 0.5 ($\sim 10 \text{ \AA}^2$ for ePSA). Solubility measured as thermodynamic solubility has greater experimental variability and is more difficult to predict, with R^2 typically below 0.5 and RMSE closer to 0.8 (decadic logarithmic or $\log_{10}$ scale) (Oprisiu and Winiwarter 2020).

In vitro ADME models

Clearance models

Pharmacological and toxicological profiles of compounds are significantly influenced by their metabolic stabilities, which can be investigated *in vitro* in cell fractions, e.g. liver microsomes, or hepatocytes from relevant species. In these assays compound depletion over time is measured and intrinsic clearance (CL_{int}) determined (Sohlenius-Sternbeck et al. 2010). Assays utilising human liver microsomes (HLM CL_{int}) and human/rat/dog hepatocytes (HH CL_{int}/RH CL_{int}/DH CL_{int}) are regularly used at AstraZeneca. HLM CL_{int} and RH CL_{int} are first-line assays usually run for all new compounds, whereas HH CL_{int} and DH CL_{int} are measured only for compounds progressed further. Over the years, we have developed *in silico* models for each of the assays mentioned above. Again, these are SVM models based on signature descriptors and implemented through the CPSign conformal prediction framework (Norinder et al. 2014; CPSign). The output of the CL_{int} models correspond to the predictions in both actual and $\log_{10}$ scale along with the estimated uncertainties reported as standard deviations.

Recently, we have invested efforts to train a multi-task convolutional neural network model for predicting human, rat, and dog clearances based on molecular images (Martínez Mora et al. 2022). The performances of the multi-task model were nearly in line with or better than the single task SVM models. However, the real benefit of a multi-task model comes from the fact that endpoints with a limited number of data points (e.g. HH CL_{int} and DH CL_{int}) can learn from other related endpoints with larger amounts of data (e.g. HLM CL_{int} and RH CL_{int}). In addition, we explored several model interpretability frameworks which build upon the multi-task model and demonstrated their potential for guiding medicinal chemistry design using experimentally confirmed products of metabolism obtained through the respective clearance experiments (Martínez Mora et al. 2022).

Caco-2 models

Two *in silico* models to predict compound transport across a human intestinal epithelial (Caco-2) cell layer are currently in place at AstraZeneca, Caco-2 intrinsic apparent permeability (P_{app}) and Caco-2 efflux ratio. The Caco-2 intrinsic P_{app} (intr-Caco2) model utilises intrinsic permeability measurements which estimate the inherent passive permeability of a compound across Caco-2 cells in apical to basolateral (A to B)

direction. Addition of an optimised drug cocktail, aimed at inhibiting the major drug transporters in the Caco-2 cell line, multidrug resistance gene 1 (MDR1, P-gp), multidrug resistance-associated protein 2 (MRP2) and breast cancer resistance protein (BCRP), ensures that active transport is precluded, which was found to be correlating with the fraction absorbed of orally administered drugs (Fredlund et al. 2017). The use of a pH gradient of 6.5 at the apical side and 7.4 at the basolateral side mimics the situation in the intestine (Fredlund et al. 2017). The dynamic experimental range of the model is considered to be between $0.01 \times 10^{-6} \text{ cm/sec}$ and $100 \times 10^{-6} \text{ cm/sec}$ ($\log P_{app}$ between -2 and 2) and general thresholds for estimating low and high intrinsic permeability are the following:

respectively.

For most projects, a permeability decision tree based on this model is a useful guide for compound progression during the lead identification and optimisation stages: if a compound is predicted with P_{app} above $5 \times 10^{-6} \text{ cm/sec}$ then it can be promoted further along the pipeline without an explicit measurement, since permeability is likely not to be a stopper in such case (Fredlund et al. 2017). Based on our earlier evaluation, compounds with a predicted P_{app} above $5 \times 10^{-6} \text{ cm/sec}$ show experimental values above $2.1 \times 10^{-6} \text{ cm/sec}$, which as mentioned earlier corresponds to more than 80% fraction absorbed (Oprisiu and Winiwarter 2020). The model itself uses a random forest algorithm (Breiman 2001) trained on physicochemical descriptors calculated by ACD (ACD/Labs Software; Advanced Chemistry Development 2015), Biobyte cLogP (ClogP, version 4.3; Pomona College and BioByte), and OESelma software (Linusson et al. 2001).

The Caco-2 efflux ratio model is based on the Caco-2 efflux assay which measures the P_{app} over the Caco-2 cell layer in both directions with no pH gradient, i.e. pH is 7.4 on both sides. The $P_{app}(B-A)/P_{app}(A-B)$ ratio is calculated to estimate the likelihood of a chemical compound to be a transporter substrate. The Caco-2 cell system contains several transporter families, including ABC efflux transporters, such as P-gp (MDR1), BCRP, and MRP2. In our experiments, an efflux ratio (ER) less than 2 indicated an unlikely substrate whereas the $ER > 2$ indicated a likely substrate, which could be of interest both for absorption or for crossing other membranes such as the blood brain barrier. An additional threshold at $ER = 7.5$ was recently suggested internally as a cue to instigate further studies to elucidate which particular efflux properties are limiting the compound's absorption potential. Similar to the intrCaco2 model, the Caco-2 efflux ratio model utilises a random forest approach (Breiman 2001) and was built on physicochemical descriptors calculated via ACD (ACD/Labs Software; Advanced Chemistry Development 2015), Biobyte cLogP (ClogP, version 4.3; Pomona College and BioByte), and OESelma software (Linusson et al. 2001).

Plasma protein binding models

Both PK and PD properties of drugs are influenced by the extent of plasma protein binding, given that only the fraction

unbound (or 'free') can effectively pass across the cell membranes, undergo metabolic transformations and kidney-mediated clearance, and manifest the intended biological effect (Fessey et al. 2006). Thus, protein binding in plasma presents a critical parameter to estimate, for example, *in vitro*-to-*in vivo* scaling of metabolic clearance or free binding affinity in cell assays. Typically, drugs bind reversibly to plasma proteins, such as albumin, α 1-acid glycoprotein, and lipoproteins, with different affinities and kinetics. For instance, highly bound drugs are largely retained in plasma which consequently restricts their tissue distribution, decreases their metabolic conversion and clearance, prolongs their half-lives, and limits their penetration into the brain (Fessey et al. 2006). Routinely, the degree to which drugs undergo plasma protein binding is measured across several different species, which expands the potential for the application of ML approaches. In total, [REDACTED]

[REDACTED], and binding to foetal calf serum proteins (fcsPPB). Commonly, the measurements are obtained using equilibrium dialysis at pH of 7.4 and 37°C with an incubation time over 18h, followed by liquid chromatography mass spectrometry detection (Wan and Rehngren 2006). The dynamic experimental range is considered to be between 10% and 99.95% bound, or logK between -0.95 and 3.30, where:

$$\log K = \log \frac{\text{fraction unbound}}{\text{fraction bound}}$$

Both logK and percent unbound (or percent free) are retrieved as the model output. All PPB models utilise the SVM algorithm (Vapnik 2000) with a linear kernel and are trained on signature descriptors (Faulon et al. 2003) implemented via the CPSign (Alvarsson et al. 2014; CPSign) conformal prediction framework.

In addition to PPB models, we also estimate fraction unbound in rat hepatocytes (RH fuinc) to understand binding in the *in vitro* system and, is used in *in vitro*-to-*in vivo* scaling of metabolic clearances. The ML model for RH fuinc also relies on a conformal prediction framework and outputs the logged ratios of bound to free % (log K).

Performances of *in vitro* ADME models

Clearance models show lower R^2 ($R^2 \sim 0.3$ - 0.4), when compared to the physicochemical models, however RMSE values are often similar (RMSE < 0.5, $\log_{10}$ scale). While the models may not be perfectly able to rank compounds correctly as seen by the low R^2 , they are useful to select compounds for synthesis that have a good chance of sufficient metabolic stability. Both the *intrCaco2* and the *efflux* models give sufficiently low RMSEs (RMSE ~ 0.5) to be of help at the design stage.

PPB models show again good overall useability, with hPPB, where most data is available, having the best performance (RMSE < 0.5 and $R^2 = 0.7$). This data indicates that a large enough dataset is, in this case, key to a good performance (Oprisiu and Winiwarer 2020).

Predicting *in vivo* pharmacokinetics

In vitro assays are only surrogates for *in vivo* PK parameters and, like most *in vitro* experiments, do not adequately capture all the mechanisms of metabolism and clearance that can be observed in a whole organism. Therefore, it has been a long-standing desire to be able to predict the outcome of *in vivo* PK experiments directly from chemical structure so that these could be used to design molecules with the optimal PK profiles. However, access to sufficiently large data sets of *in vivo* data has precluded the development of ML models until now. The recent rising interest in using artificial intelligence has created an increasing demand for data to be made accessible for these approaches and so there is now an opportunity to explore the use of ML in predicting *in vivo* PK parameters.

Work published by Sia Low et al. (Low et al. 2014) suggests that using a combination of chemical descriptors and biological assays can improve the prediction of *in vivo* effects. Therefore, we explored combining the traditional approaches using chemical structure descriptors with *in vitro* assay results in order to predict *in vivo* PK profiles. Missing experimental values were substituted with predicted results (Figure 3).

In vivo pharmacokinetic models

We have established a suite of predictive models for *in vivo* animal and human PK, enabling prediction from chemical structure early in drug discovery, and therefore offering opportunities to guide design and prioritisation of molecules based on relevant *in vivo* properties and, ultimately, predicting human PK at the point of design.

In vivo rat PK model

We have previously reported on our efforts to model *in vivo* rat PK parameters (IV and PO administration) and concentration-time PK profiles (Obrezanova et al. 2022). The models were based on a dataset of over 3,000 diverse compounds and used features including molecular chemical structure descriptors and measured *in vitro* (or predicted) ADME properties, and utilised multi-task graph convolutional neural networks and traditional ML methods. The deep learning multi-task algorithms showed performance equivalent to classical ML. In addition, the CL model demonstrated higher predictive accuracy than the traditional *in vitro*-to-*in vivo* extrapolation (IVIVE) approach. The multi-task Chemprop (Yang et al. 2019; Heid et al. 2024) model achieving coefficient of determination $R^2 = 0.55$ (Root Mean Squared Error, RMSE = 0.46, 65% of compounds within 2-fold error) for F and $R^2 = 0.63$ (RMSE = 0.26, 78% within 2-fold error) for CL has been used in our design-make-analyze-test (DMTA) cycle to guide compound prioritisation alongside other ADME and safety predictive models. The multi-task model was later rebuilt on a much larger dataset of 24,000 compounds including PROTACs and a wider range of doses, including high doses exceeding 1000 mg/kg. The model was trained on a combination of compound-dose pairs. Figure 4 shows that performance of the extended rat PK model has improved in

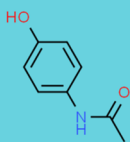


Figure 3. The AstraZeneca approach to predicting *in vivo* PK parameters using a combination of chemical structure and experimental results, supplementing these with predicted values where they are absent.

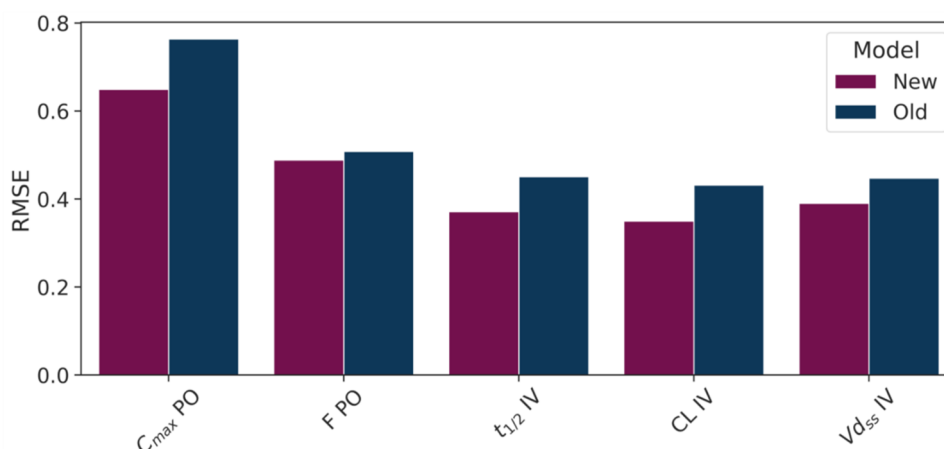


Figure 4. Performance of extended (New) *in vivo* rat PK model (trained on $N=24,376$ compounds) versus (Old) model (trained on 4,308 compounds). RMSE evaluated on a common test set is shown for five PK parameters.

comparison to the previous model. The model is rebuilt every two months; [Figure 5](#) shows the performance of the model for CL and F over time and demonstrates that there are variations in the model's performance, particularly for the F parameter, which is more difficult to predict as it is related to a compound's absorption and clearance properties.

Figure 6 demonstrates the effect of regular model updates for a series of acidic or zwitterionic compounds, which were found to show hepatic uptake. The initial set of compounds (light grey symbols) were not well predicted, but after including first compounds in the model training set better results were achieved for late-stage compounds (black symbols).

In vivo mouse PK model

Dataset, features and modelling approach

Mouse is a common species for preclinical PD models and also of interest for *in vivo* PK experiments, at AstraZeneca often used for PROTACs discovery programs. The *in vivo* mouse PK model was built utilising a multi-task graph convolutional neural network approach Chemprop (Heid et al.

2024) following the same approaches as applied to *in vivo* rat PK modelling (Obrezanova et al. 2022).

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The input features of the model included dose, experimentally obtained ADME and physicochemical properties, 1D and 2D OESelma molecular descriptors (Linusson et al. 2001;

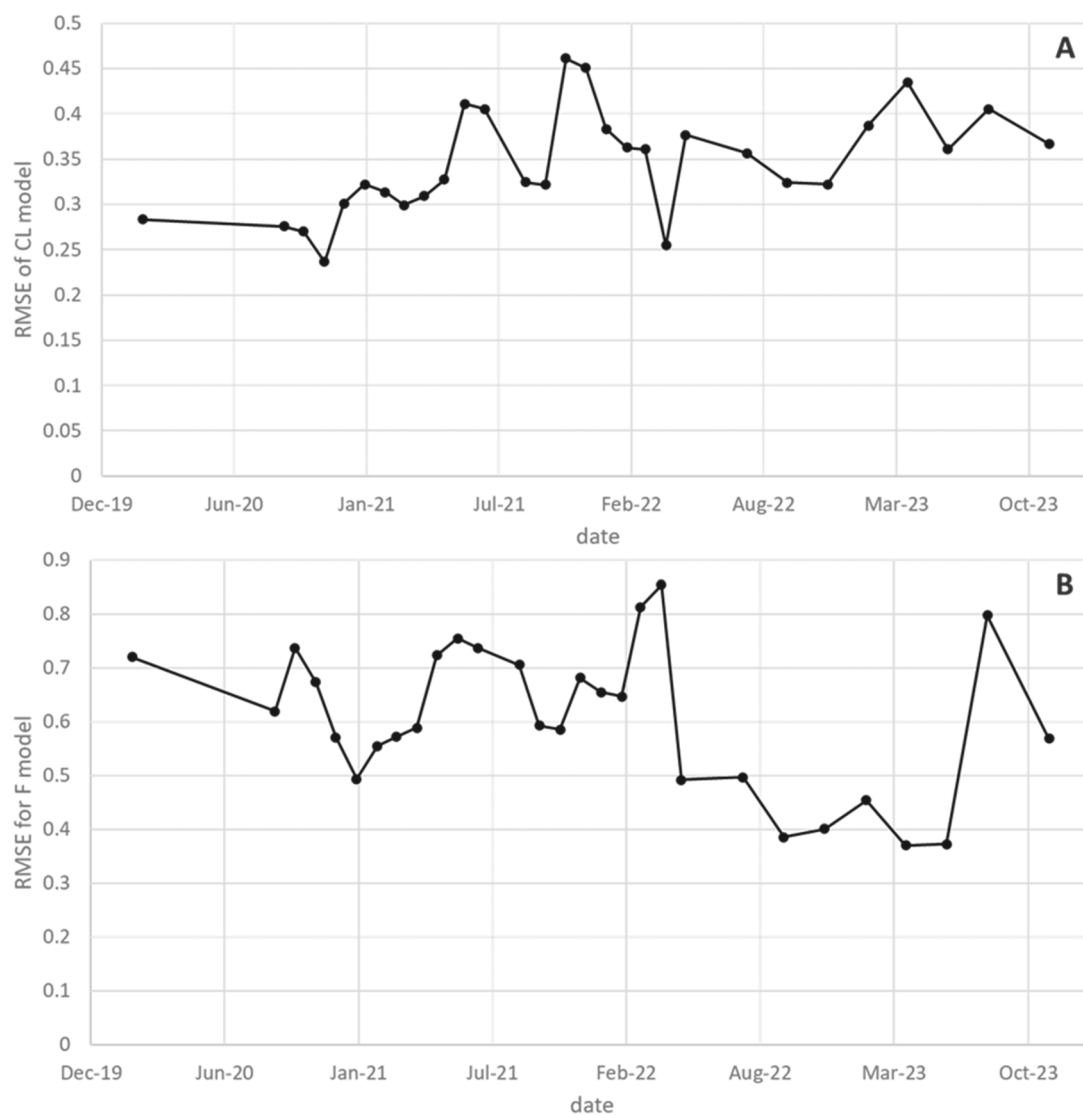


Figure 5. Performance of *in vivo* rat PK model over time. RMSE of the model is evaluated on temporal test sets for the (A) CL model and (B) F model. RMSE is calculated in log-transformed space.

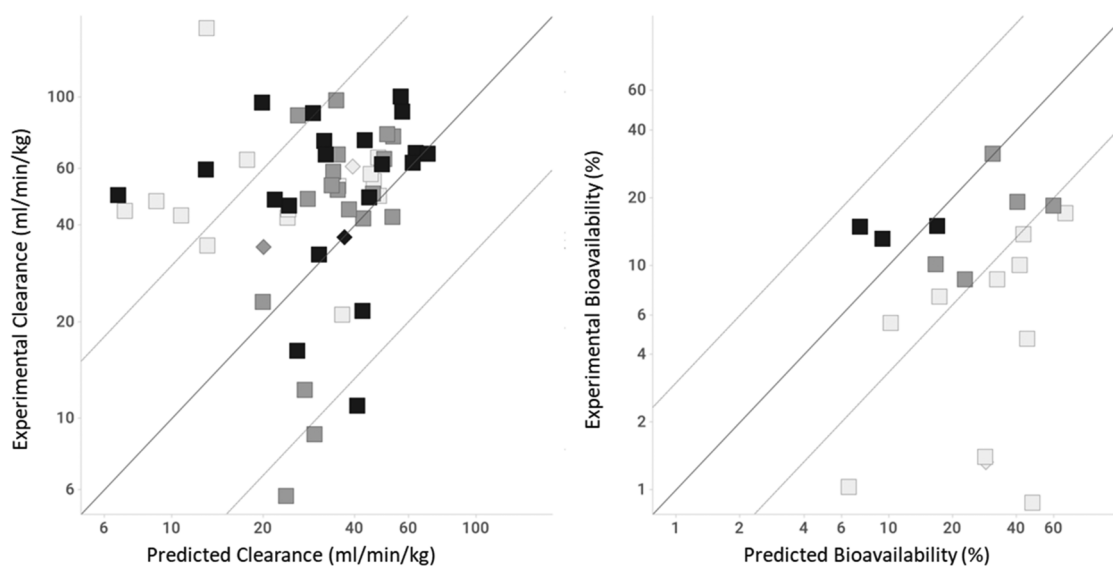


Figure 6. Left panel: Rat clearance ($n=56$), Right panel: Rat oral bioavailability ($n=19$); light grey symbols: early-stage compounds, dark grey symbols: middle-stage compounds, black symbols: late-stage compounds; squares: acids, diamonds: zwitterions; dark line: unity, light line(s): 3-fold.

Obrezanova et al. 2022) and molecular representations encoded by a Chemprop graph convolutional network (Yang et al. 2019). The OESelma descriptors comprise around 100 common 1D and 2D molecular descriptors related to physicochemical properties, such as size, ring structure, flexibility, etc.

These *in vitro* properties are commonly measured early and are available for many compounds. Missing *in vitro* data points were imputed by corresponding *in silico* predictions (see Sections Predicting physicochemical properties and *In vitro* ADME models).

The dataset was divided into training (90%) and test (10%) subsets based on random, scaffold-balanced, and temporal splitting strategies. The models were built on the training sets using five-fold cross validation (scaffold-balanced approach) to optimise hyperparameters of the model and were validated on the test sets.

Performance of mouse PK model

The performance statistics for the mouse PK models depend on the chosen splitting strategy to obtain the test set, as illustrated in Figure 7. The temporal set split represents the most realistic and challenging scenario for prediction because with time chemical series start to diverge from the initial chemical space of the model. For the majority of PK parameters, RMSE on the temporal set split is higher than RMSE on random or scaffold-balanced test sets. For most PK parameters, including $V_{d_{ss}}$ IV and F, the RMSE of the model compares favourably with the experimental variability. This is not the case for CL IV, where experimental variability is much lower than the RMSE of the prediction. The experimental variability for each PK parameter is defined as the 95% quantile of the distribution of standard deviations between compound replicates.

Tables 1 and 2 provide the performance of the final mouse PK model evaluated on the temporal test set and on the subset of PROTACs, respectively. Model performance on the PROTACs subset is better than on the overall test set. The mouse model performance on CL IV and $V_{d_{ss}}$ IV is lower than that of the rat PK model (see Section *In vivo* rat PK model), but the performance for F is comparable. The observed versus predicted values for F are shown in Figure 8, for the temporal test set and for the PROTACs subset. The accuracy of F model was deemed to be sufficient, especially for PROTACs. The model for F was also evaluated as a classification model with two thresholds: 10% and 30%. For the 10% threshold, the F model achieved accuracies of 71% and 76% on the test set and the PROTACs subset, respectively. For the threshold of 30%, the model achieved accuracies of 70% and 84%, respectively.

In vivo dog PK model

Similar to the described rodent PK models, the dog PK model implements the multi-task directed message-passing neural network based on the Chemprop framework (Yang et al. 2019; Heid et al. 2024). The model is trained on the combination of doses (mg/kg), Chemprop-derived chemical graph encodings, 1D and 2D descriptors, as well as *in vitro* (or *in silico* predictions were unavailable) ADME and physicochemical properties. In total, 5,083 measured compound-dose pairs, corresponding to 2,717 unique compound structures, were applied to build and evaluate the multi-task model for seven PO/IV parameters: F (classification outcome with a threshold of 30%: $F \leq 30\%$, low F; $F > 30\%$, high F), CL IV (ml/min/kg), AUC PO ($\mu\text{M} \cdot \text{h}$), C_{max} PO and IV (μM), $t_{1/2}$ (h), and $V_{d_{ss}}$ IV (L/kg). Experimental outcomes for ten *in vitro* physicochemical and ADME assays were employed as input features to the model, where in the cases of missing values (not measured or beyond the limit of detection) the corresponding *in silico* models were considered to impute the prediction values (~31% of the data points were unpopulated). During model benchmarking, we noticed little or no performance difference between using a combination of *in vitro* and *in silico* values

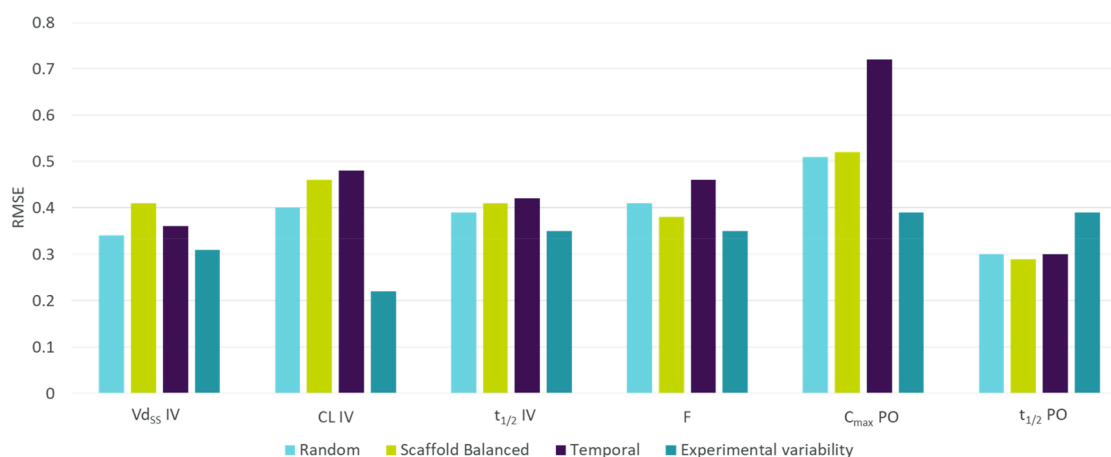


Figure 7. Mouse PK model performance for six PK parameters evaluated on different test sets based on random, scaffold-balanced and temporal splitting strategies. RMSE (in log-transformed space) is shown as well as experimental data variability. The models are based on a smaller set of compounds (3,191 compounds overall, 10% in the test sets).

PK parameter	N	R ²	RMSE	% within 2-fold error	% within 3-fold error	Experimental variability
CL IV	197	0.38	0.42	54	79	0.22
C _{max} PO	172	0.44	0.59	47	60	0.39
F	172	0.43	0.44	57	78	0.35
Vd _{ss} IV	197	0.48	0.37	68	88	0.31
t _{1/2} IV	197	0.36	0.38	65	81	0.35
t _{1/2} PO	171	0.26	0.29	79	90	0.39

PK parameter	N	R ²	RMSE	% within 2-fold error	% within 3-fold error	Experimental variability
CL IV	63	0.24	0.31	63	90	0.22
C _{max} PO	55	0.35	0.43	65	76	0.39
F	55	0.67	0.33	67	87	0.35
Vd _{ss} IV	63	0.66	0.24	81	92	0.31
t _{1/2} IV	63	0.53	0.26	78	90	0.35
t _{1/2} PO	55	0.079	0.26	84	93	0.39

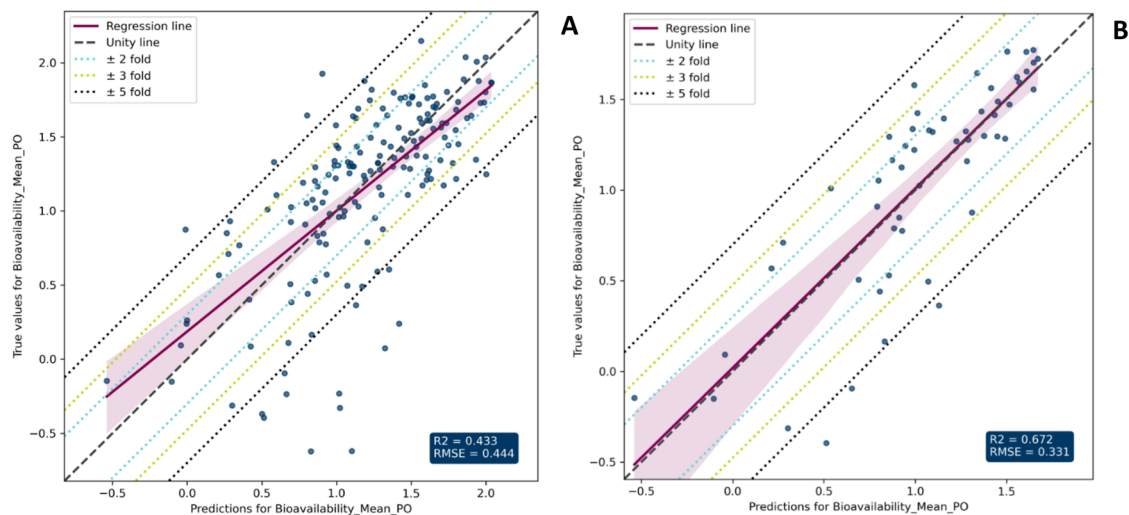


Figure 8. Observed versus predicted values for log₁₀(F) on the temporal test set (A) and on the PROTACs subset (B).

and only *in silico* estimates for the major PK parameters (F, CL IV, and Vd_{ss} IV).

Following assays and their corresponding models were used:

The model performance was evaluated using different data splitting strategies (training-test set ratio set to 90:10), including stratified, random, scaffold, singleton, and temporal splits. For the latter, the test set consisted of the compounds measured after 2014, which posed a difficult model validation scenario. However, this was to be expected given that dog *in vivo* PK experiments are only rarely performed, typically for 1-2 project compounds where dog acts as a second pre-clinical species (after rat/mouse). Therefore, we considered the singleton split (test set compounds having no available structural analogues within the data, i.e. containing a unique scaffold) to evaluate the model performance and concluded decent accuracy where the RMSE values were approaching the estimated experimental assay variability (Table 3).

Table 3. Performance statistics for dog PK model. Shown are R² and RMSE (in log₁₀) values for six dog *in vivo* PK parameters regression models as evaluated on the singleton split. For F the regression model predictions were converted to a classification outcome, captured here by the Mathews correlation coefficient (MCC) value. Experimental errors (log₁₀) are provided as well.

Parameter	R ²	RMSE	MCC (thr. 30%)	Experimental variability
CL IV	0.53	0.34		0.25
C _{max} IV	0.59	0.35		0.32
t _{1/2} IV	0.27	0.39		0.34
Vd _{ss} IV	0.50	0.34		0.29
AUC PO	0.49	0.60		0.43
F			0.38	0.42
C _{max} PO	0.51	0.56		0.47

In vivo human PK model

Ultimately, the machine learning models for the prediction of pre-clinical *in vivo* PK serve to aid in estimating drug exposure and safety in the human patient population. We have

(Miljković et al.

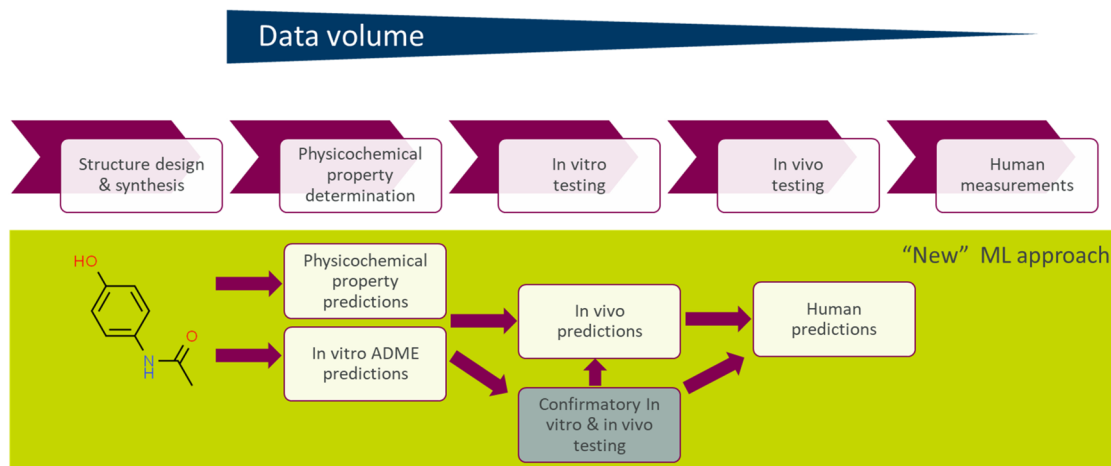


Figure 9. The future approach to predicting human PK parameters using experimental data where available or predicted properties where it is missing.

2021). In total, [REDACTED]

[REDACTED]

[REDACTED] These models were trained on available compound doses (mg), 1D and 2D descriptors, and in-house ADME/PK model predictions. Following a rigorous validation protocol, [REDACTED]

[REDACTED]

[REDACTED] Although the ML models showed slightly lower performance compared to DMPK estimates, their early application as a tool complementary to the resource intensive DMPK calculations, particularly before a compound is synthesised and experimentally evaluated, is pivotal to the DMTA cycle acceleration and the timely delivery of future clinical candidates.

Conclusions

Being able to accurately predict the *in vivo* PK properties of a compound at the point of design will ultimately lead to improved efficiency in the drug discovery process. However, the aim is to be able to predict the human PK profile of novel drug candidates. We have shown that using chemical structures, predictions of physicochemical properties and *in vitro* assays as inputs into the models for *in vivo* PK yields greater predictive performance for preclinical species. Similarly, [REDACTED]

[REDACTED]

[REDACTED] Allowing the models to accept experimentally derived properties where they have been measured or calculate them where they do not exist as shown in Figure 9, is expected to yield improved performance in predicting human PK as molecules progress through the discovery process, but still with the ability to predict these profiles at the point of design.

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ORCID

Nigel Greene  <http://orcid.org/0000-0003-0433-4596>

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